

# #Aviation Meteorology



## *Chapter Nine: Coding, Decoding, and Reporting*

---



### **Objectives**

- To Be aware of the Weather report and forecast
- To Know International met codes
- To Describe the Meteorological broadcasts for aviation

# Code, Decode, and Report

- Weather reports and forecasts are critical components of aviation weather services.
- These services provide pilots with important information about the weather conditions at their departure airport, along their route of flight, and at their destination airport.
- In order to communicate this information effectively, weather reports and forecasts use a standardized coding system that is easy to understand and decode.

- **Coding:** Weather reports and forecasts use a coding system to communicate weather information in a concise and standardized format.
- This coding system uses *abbreviations and symbols* to represent weather phenomena, such as cloud cover, precipitation, and visibility.
- For example, the abbreviation "SCT" is used to represent "scattered" clouds, while the symbol "RA" is used to represent "rain."

- **Decoding:** To decode a weather report or forecast, pilots must be familiar with the coding system and *understand the meanings of the abbreviations and symbols.*
- For example, a report that includes the abbreviation "BKN" indicates "broken" clouds, while a report that includes the symbol "SN" indicates "snow."
- Pilots must also be able to interpret the information provided in the report, such as wind speed and direction, temperature, and altimeter setting.

- **Reporting:** Weather reports and forecasts are typically issued by meteorological authorities or organizations.
- These reports may be issued periodically, such as hourly or every six hours, depending on the location and weather conditions.
- Weather reports and forecasts are disseminated to pilots through a variety of channels, such as radio broadcasts, ATIS systems, and pre-flight weather briefings.
- Pilots are required to use this information to make informed decisions about their flight, including route selection, altitude, and fuel requirements.

- TAFOR, METAR, SPECI, TREND, SIGMET, AIREPs, RAREPs, ROFOR SYNOP are all international meteorological codes used to report weather information in the aviation industry.
- All of these codes are standardized by the International Civil Aviation Organization (ICAO), and are used by aviation weather specialists around the world to report and interpret weather information.

➤ By using these codes, pilots and air traffic controllers can make informed decisions about flight operations, including route planning, altitude selection, and takeoff and landing operations, while ensuring the safety and efficiency of aviation operations in a range of weather conditions.



# **Weather Report And Forecast:- International Met Codes**

- **TAF (Terminal Aerodrome Forecast):** A TAF is a forecast of expected weather conditions at an aerodrome for a period of 24 to 30 hours.
- It is typically issued four times a day, and includes information about expected changes in **wind direction and speed, visibility, cloud cover, and precipitation.**
- TAFs are used by pilots and air traffic controllers to plan flight operations and make informed decisions about airport operations.

## Cont...

➤ **Example TAF:** TAF VHHH 130800Z 1309/1415 12010KT 8000  
SCT018 TEMPO 1312/1316 4000 -RA BECMG 1315/1317  
22008KT PROB30 TEMPO 1400/1404 4000 SHRA BKN012

Decoding:

- TAF VHHH: The TAFOR is for Hong Kong International Airport (VHHH).
- 130800Z: The TAFOR was issued on the 13th day of the month at 0800Z (UTC).
- 1309/1415: The TAFOR is valid from 0900Z on the 13th to 1500Z on the 14th.
- 12010KT: Wind from 120 degrees at 10 knots.
- 8000: Visibility of 8000 meters.
- SCT018: Scattered clouds at 1800 feet above ground level.
- TEMPO 1312/1316: Temporary fluctuations in the weather are expected between 1200Z and 1600Z on the 13th.
- 4000 -RA: Visibility of 4000 meters with light rain.
- BECMG 1315/1317: Gradual change in the weather is expected between 1500Z and 1700Z on the 13th.
- 22008KT: Wind from 220 degrees at 8 knots.
- PROB30: There is a 30% probability of the following event occurring.
- TEMPO 1400/1404: Temporary fluctuations in the weather are expected between 0000Z and 0400Z on the 14th.
- 4000 SHRA BKN012: Visibility of 4000 meters with showers of rain and broken clouds at 1200 feet above ground level.

Cont...

➤ **METAR (Meteorological Aerodrome Report):** A

METAR is a weather report that provides **current weather conditions** at an aerodrome, including **temperature, dew point, wind direction and speed, visibility, cloud cover, and precipitation.**

➤ METARs are typically issued hourly, or more frequently if conditions change significantly, and are used by pilots and air traffic controllers to make informed decisions about airport operations.

## Cont...

➤ **Example METAR:** METAR KMSP 131453Z 16007KT 10SM  
OVC060 10/06 A2999 RMK AO2 SLP155 T01000056 51004

Decoding:

- METAR KMSP: The report is for Minneapolis-St. Paul International Airport (KMSP).
- 131453Z: The report was issued on the 13th day of the month at 1453Z (UTC).
- 16007KT: Wind from 160 degrees at 7 knots.
- 10SM: Visibility of 10 statute miles.
- OVC060: Overcast clouds at 6000 feet above ground level.
- 10/06: Temperature of 10 degrees Celsius and dew point of 6 degrees Celsius.
- A2999: Altimeter setting of 29.99 inches of mercury.
- RMK AO2 SLP155 T01000056 51004: Remarks include the presence of an automated weather station (AO2), sea-level pressure of 1015.5 hectopascals (SLP155), a temperature of 10.0 degrees Celsius (T0100), a dew point of 5.6 degrees Celsius (0056), and a 3-hour pressure change of -0.04 inches of mercury (51004).

Cont...

- **SPECI (Aviation Selected Special Weather Report):** A SPECI is a special weather report that provides updated information about significant changes in weather conditions at an aerodrome, such as sudden changes in wind direction or speed, visibility, or precipitation.
- SPECIs are typically issued between regular METAR reports when conditions change significantly, and are used by pilots and air traffic controllers to make informed decisions about airport operations.

Cont...

➤ **Example SPECI:** SPECI KJFK 131540Z 06012KT 1/2SM  
R04R/1200V1600FT SN FZFG VV005 M03/M04 A2984 RMK  
SNB39 SLP105

Decoding:

- This SPECI report was taken at 1540 UTC on April 13th at John F. Kennedy International Airport (KJFK).
- The wind was blowing from 060 degrees at 12 knots,
- the visibility was 1/2 statute mile,
- the runway visual range for runway 4R was between 1200 and 1600 feet,
- there was snow and freezing fog with a vertical visibility of 500 feet,
- the temperature was -3 degrees Celsius, the dew point was -4 degrees Celsius,
- the altimeter setting was 29.84 inches of mercury,
- there was a snow depth of 39 inches, and
- the sea level pressure was 1010.5 hPa.

- SPECI: It indicates that this is a special weather report, typically issued when there are significant changes in weather conditions between routine METAR reports.
- KJFK: The ICAO code for John F. Kennedy International Airport.
- 131540Z: The date and time of the report in UTC. In this case, it is the 13th day of the month at 1540 Zulu time.
- 06012KT: Wind information. The wind is coming from 060 degrees at a speed of 12 knots.
- 1/2SM: Visibility is 0.5 statute miles.
- R04R/1200V1600FT: Runway visual range for Runway 04R. The visibility is varying between 1200 and 1600 feet.
- SN FZFG: Snow and freezing fog are present.
- VV005: Vertical visibility is 500 feet. This indicates a cloud layer with a base at 500 feet above ground level.
- M03/M04: Temperature is -3°C, and dew point is -4°C.
- A2984: Altimeter setting is 29.84 inches of mercury.
- RMK SNB39 SLP105: Additional remarks, including the snow depth (SNB39) and sea-level pressure (SLP105).



Cont...

- **TREND:** The TREND code is used to provide information on the expected changes in weather conditions at an aerodrome.
- The code includes information on the expected changes in wind direction, wind speed, visibility, cloud cover, and precipitation.

Cont...

- **Example TREND:** TREND TEMPO FM1530 TL1600 3000 -RA  
BR SCT008 BKN012

Decoding:

- From 1530 UTC until 1600 UTC,
- temporary fluctuations are expected with visibility of 3000 meters,
- light rain and mist,
- scattered clouds at 800 feet, and
- broken clouds at 1200 feet.

Cont...

- **SIGMET:** The SIGMET code is used to indicate the presence of **severe or extreme weather conditions**, such as **thunderstorms**, **turbulence**, **icing**, or **volcanic ash clouds**, which may affect flight safety.
- SIGMET messages are issued by meteorological agencies and are broadcast to all aircraft in the affected area.

Cont...

- **Example SIGMET:** WSSS SIGMET 02 VALID 130700/131100  
WSSS- SINGAPORE FIR OBS VA EMISSION TO FL100

Decoding:

- A volcanic ash cloud has been observed within the Singapore Flight Information Region (FIR),
- which may pose a hazard to aircraft up to 10,000 feet (100\*100)above sea level.
- The SIGMET is valid from 0700 UTC to 1100 UTC.

Cont...

- **SYNOP:** The SYNOP code is used to report weather observations *from land-based weather stations.*
- SYNOP reports include information on temperature, dew point, wind speed and direction, pressure, precipitation, cloud cover, and visibility.

# **Meteorological broadcasts for aviation**

- Meteorological broadcasts for aviation are essential in providing pilots with weather information to ensure safe and efficient flight operations.
- Here are some of the most common types of meteorological broadcasts for aviation:
- **VOLMET:** VOLMET broadcasts provide weather information over designated frequencies for flights operating within a particular area or route.
- These broadcasts typically include information on winds, temperature, dew point, visibility, and significant weather.

Cont...

- **Example** VOLMET : VOLMET EIDW 101100Z 30006KT 9999  
SCT016 BKN040 07/03 Q1014 NOSIG

Decoding:

- This VOLMET broadcast was issued on the 10th of the month at 1100 UTC for Dublin Airport (EIDW).
- The wind was blowing from 300 degrees at 6 knots,
- the visibility was 9999 meters,
- there were scattered clouds at 1600 feet and broken clouds at 4000 feet,
- the temperature was 7 degrees Celsius, the dew point was 3 degrees Celsius,
- the altimeter setting was 1014 hPa, and
- there were no significant changes expected.



## Cont...

- **ATIS:** The Automatic Terminal Information Service (ATIS) is a continuous broadcast of meteorological information that is updated regularly.
- ATIS messages are transmitted on a specific frequency and provide information on weather conditions at the airport, such as wind direction and speed, temperature, dew point, visibility, cloud coverage, and runway conditions.

## Cont...

- **Example ATIS:** KJFK 101350Z 30007KT 10SM BKN015 BKN025 OVC035 05/03 A3012 RMK AO2 SLP203 T00500028

### Decoding:

- This ATIS broadcast was issued on the 10th of the month at 1350 UTC for John F. Kennedy International Airport (KJFK).
- The wind was blowing from 300 degrees at 7 knots,
- the visibility was 10 statute miles,
- there were broken clouds at 1500 feet,
- broken clouds at 2500 feet, and overcast clouds at 3500 feet,
- the temperature was 5 degrees Celsius, the dew point was 3 degrees Celsius,
- the altimeter setting was 30.12 inches of mercury,
- there were automated weather observation systems present,
- the sea level pressure was 1020.3 hPa, and
- the temperature and dew point were 5 and 2.8 degrees Celsius, respectively.

Cont...

- **HFVOLMET:** HFVOLMET is similar to VOLMET but is broadcast over high-frequency (HF) radio frequencies, which allows pilots to receive weather information over long distances.
- These broadcasts are particularly useful for flights operating over remote or oceanic areas.

## Cont...

- **Example HFVOLMET:** KJFK 101350Z 30007KT 10SM BKN015 BKN025 OVC035 05/03 A3012 RMK AO2 SLP203 T00500028

### Decoding:

- This HFVOLMET broadcast was issued between 1200 and 1800 UTC for Lima, Peru.
- The weather information provided includes the following:
- the wind direction and speed at 8891 meters,
- the temperature at 13205 meters, the temperature and dew point at 10118 meters,
- the wind direction and speed at 34006 meters,
- the geopotential height at 41162 meters,
- the wind direction and speed at 61401 meters, and
- the temperature and dew point at 91255 meters.

## Weather

Weather is indicated by up to 3 groups comprising symbols and letters from the following:

|                   |                               |                                  |                      |
|-------------------|-------------------------------|----------------------------------|----------------------|
| - = slight        | + = Heavy                     | BC = Patches                     | BL = Blowing         |
| BR = Mist         | DR = Low Drifting             | DS = Dust Storm                  | DU = Widespread Dust |
| DZ = Drizzle      | FG = Fog                      | FC = Funnel Cloud (e.g. Tornado) | FU = Smoke           |
| FZ = Freezing     | GR = Hail                     | GS = Small Hail                  | HZ = Haze            |
| IC = Ice Crystals | MI = Shallow                  | PL = Ice Pellets                 | PO = Dust Devils     |
| RA = Rain         | SA = Sand                     | SG = Snow Grains                 | SH = Shower          |
| SN = Snow         | SQ = Squall                   | SS = Sandstorm                   | TS = Thunderstorm    |
| VA = Volcanic Ash | VC = In the vicinity (nearby) | UP = Unidentified Precipitation  | RE = Recent          |

## Cloud

Cloud amount is reported as **FEW** = few (1-2 oktas), **SCT** = scattered (3-4 oktas), **BKN** = broken (5-7 oktas) or **OVC** = overcast (8 oktas), followed by the height of the cloud base in hundreds of feet above aerodrome elevation e.g. SCT018 = scattered clouds at 1800 ft. Note that in area and route forecasts, heights are AMSL or standard pressure altitudes.

Selection of cloud layers reported is made as follows:

- The lowest Layer;
- The next lowest layer of **SCT** or more;
- The next higher layer of **BKN** or more;
- Significant convective cloud (**CB** or **TCU**) if not already reported.

....END

Thanks !